

MINI ASSESSMENT - LAB REPORT

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1. AIM

To create and implement a simple rule-based conversational chatbot in Python that accepts user messages, identifies basic intents through conditional statements, and produces suitable responses during an interactive session.

2. PROCEDURE

1. Start a continuous `while True` loop so that the chatbot can interact with the user repeatedly.
2. Read each message from the keyboard using Python's `input()` function.
3. Clean the entered message using `strip()` and `lower()` so that extra spaces and capitalization do not affect matching.
4. Check whether the user entered a termination word such as `bye`, `exit`, or `quit`.
5. Use an `if-elif-else` decision structure to recognize greetings, status questions, identity questions, and help requests.
6. Use the final `else` branch to provide a friendly response when the input does not match any supported rule.

3. FLOWCHART

```
[START] → [Show Welcome Message] → [while True Loop] → [Read User Input]
      ↓
    [strip() + lower()]
      ↓
  [Is input bye / exit / quit?]
    YES → [Goodbye Message] → [END]
    NO
      ↓
  [Greeting?] YES → [Greeting Response]
    NO → [How are you?] YES → [Status Response]
    NO → [Name?] YES → [Identity Response]
    NO → [Help?] YES → [Help Response]
    NO → [Fallback Response] → [Loop Back]
```

4. PYTHON CODE

```

def rule_based_bot():
    print("Bot: Hello! I am your Python assistant.")
    print("Type 'bye' or 'exit' whenever you want to stop.\n")

    while True:
        user_input = input("You: ").strip().lower()

        if user_input in ["bye", "exit", "quit"]:
            print("Bot: Goodbye! Have a great day!")
            break

        elif user_input in ["hello", "hi", "hey"]:
            print("Bot: Hello! How can I assist you?")

        elif "how are you" in user_input:
            print("Bot: I am doing well. Thanks for asking!")

        elif "your name" in user_input:
            print("Bot: I am a simple Python rule-based chatbot.")

        elif "help" in user_input:
            print("Bot: You can greet me, ask my status or name, "
                  "request help, or type 'bye' to exit.")

        else:
            print("Bot: Sorry, I do not recognize that request. "
                  "Try typing 'help'.")

if __name__ == "__main__":
    rule_based_bot()

```

4. PYTHON CODE (CONTINUED)

```
# The decision branches above are evaluated in order.  
# The first matching rule produces the response.  
  
# Unrecognized input is handled by the final else block.  
# The loop continues until a termination word is entered.
```

5. RESULT

```
Bot: Hello! I am your Python assistant. Type 'bye' or 'exit' whenever you want to stop.  
You: hi  
Bot: Hello! How can I assist you?  
You: how are you?  
Bot: I am doing well. Thanks for asking!  
You: what is your name?  
Bot: I am a simple Python rule-based chatbot.  
You: can you sing?  
Bot: Sorry, I do not recognize that request. Try typing 'help'.  
You: bye  
Bot: Goodbye! Have a great day!
```

6. CONCLUSION

The rule-based chatbot was successfully developed using Python conditional branching and a continuous while loop. It accepts user input, performs basic keyword-based intent matching, provides predefined responses, and terminates cleanly when an exit command is entered.